PROGRAM-3

1. Billing Process

Input Data:

- Patient ID
- Patient Name
- Consultation Charges
- Diagnostic Charges
- Treatment Charges
- Patient Category (General, Senior Citizen, Employee, Insured)

Total Bill Calculation:

$$\text{Total Charges} = \text{Consultation} + \text{Diagnostic} + \text{Treatment}$$

2. Conditional Concessions

Example policy:

Category	Concession
Senior Citizen	15%
Hospital Employee	20%
Insured Patient	10%
General Patient	0%

Discount Amount

$$\text{Discount} = \text{Total Charges} \times \frac{\text{Concession Rate}}{100}$$

Final Bill

$$\text{Final Bill} = \text{Total Charges} - \text{Discount}$$

3. Billing Constraints Validation

- Consultation charge ≥ 0
- Diagnostic charge ≥ 0
- Treatment charge ≥ 0
- Total bill must not exceed predefined system limit (if applicable)

- Patient category must be valid
- Final bill cannot be negative

4. Patient Classification

Based on final payable amount:

Final Bill Amount	Classification
Above ₹50,000	Premium Patient
₹20,000 - ₹50,000	Standard Patient
Below ₹20,000	Basic Patient

5. Algorithm

1. Read patient details and charges.
2. Calculate total charges.
3. Determine concession percentage based on category.
4. Compute discount amount.
5. Calculate final bill.
6. Validate billing constraints.
7. Classify patient according to final bill amount.
8. Display bill summary and patient classification.

Example

Input

- Consultation = ₹1,000
- Diagnostic = ₹4,000
- Treatment = ₹25,000
- Category = Senior Citizen

Calculation

- Total Charges = ₹30,000
- Concession = 15%
- Discount = ₹4,500
- Final Bill = ₹25,500

Output

- Total Bill: ₹30,000

- Discount: ₹4,500
- Final Payable Amount: ₹25,500
- Classification: **Standard Patient**

Programm:

```
    # Hospital Billing and Patient Classification


# Input patient details
patient_name = input("Enter patient name: ")
age = int(input("Enter patient age: "))

# Input charges
consultation = float(input("Enter consultation charges: "))
diagnostic = float(input("Enter diagnostic charges: "))
treatment = float(input("Enter treatment charges: "))

# Input payment details
payment_type = input("Enter payment type (Cash/Insurance): ").strip().lower()

# Validate inputs
if age <= 0 or consultation < 0 or diagnostic < 0 or treatment < 0:
    print("Invalid input. Please enter valid details.")

else:
    # Calculate total bill
    total_bill = consultation + diagnostic + treatment

    # Apply concessions
    concession = 0

    if age >= 60:
        concession = total_bill * 0.10
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elif total_bill >= 10000:
    concession = total_bill * 0.05

# Calculate final bill
final_bill = total_bill - concession

# Classify patient according to payment policy
if payment_type == "insurance":
    patient_category = "Insurance Patient"
elif payment_type == "cash":
    patient_category = "Self-Paying Patient"
else:
    patient_category = "Invalid Payment Type"

# Display bill
print("\n----- HOSPITAL BILL -----")
print("Patient Name:", patient_name)
print("Patient Age:", age)
print("Consultation Charges: Rs.", round(consultation, 2))
print("Diagnostic Charges: Rs.", round(diagnostic, 2))
print("Treatment Charges: Rs.", round(treatment, 2))
print("Total Bill: Rs.", round(total_bill, 2))
print("Concession: Rs.", round(concession, 2))
print("Final Bill: Rs.", round(final_bill, 2))
print("Patient Classification:", patient_category)
```